AN EXACT RATIONAL CERTIFICATE FOR A SPHERICAL LAUGHLIN GAP BOUND

ABSTRACT. For the fermionic spherical V_1 Hamiltonian with coefficient one on each pair projector and flux $Q = 3(N-1)$, an exact rational certificate gives a gap of at least $\frac{4380340253}{20000000000} = 0.21901701265$ for all sufficiently large N. The construction uses the Fock-space comparison of OpenAI and the enlarged certificate, product-vector averaging, and staircase allowance estimates of Gao. It changes only the pair term in Gao's rows by the rational multiplier 9999999999/10000000000. Exact finite checks are combined with a relative transfer that accommodates singular Gram matrices and a summable three-body tail. The theorem holds for every sufficiently large system size.

1. STATEMENT, CONVENTIONS, AND ATTRIBUTION

Let U_Q be the unitary spin-$Q/2$ representation of SU(2), with orthonormal lowering basis $e_0, \dots, e_Q$ and $J_- e_j = \sqrt{(j+1)(Q-j)}\, e_{j+1}$. Increasing exterior products are orthonormal. On $\bigwedge^N U_Q$, put

$$H_{N,Q} = \sum_{1 \le i < j \le N} P_{ij}^{(1)},$$

where $P_{ij}^{(1)}$ is the orthogonal pair projector onto spin $Q-1$ (relative angular momentum one). No factor depending on N or Q multiplies this sum. These are the conventions of the original spherical gap paper [2] and its stability companion [3].

Theorem 1. *There is an integer N_0 such that, for every integer $N \ge N_0$, at $Q = 3(N-1)$ the gap of $H_{N,Q}$ above the Laughlin ground-state line is at least*

$$\gamma = \frac{4380340253}{20000000000} = 0.21901701265 > 0.2190037845063196.$$

The auxiliary certificate and its transfer to all sufficiently large spheres use exact identities and inequalities.

The proof follows OpenAI's normal-ordered Fock-space comparison [2, Sections 2–6]. The enlarged integer rows, product-twirl estimate, and staircase allowance estimate are due to Gao [1]. The change of pair coefficients is stated explicitly in Section 3. No new certificate family is asserted. The conclusion concerns the unperturbed spherical Hamiltonian; no disorder threshold or additional planar conclusion is claimed here. In particular, N_0 is not numerically specified.

2. LIFTING AND SPIN IDENTITIES

Write $\mathcal{F}_Q = \bigoplus_n \bigwedge^n U_Q$. For an exterior vector v, let $B(v)$ be the adjoint of left exterior multiplication by v, and put $c_j = B(e_j)$. Thus $\{c_i, c_j^\dagger\} = \delta_{ij}$, $\{c_i, c_j\} = 0$, $\|c_j\| \le 1$, and $B(v \wedge w) = B(w)B(v)$. The lift

$$\mathcal{L}_k(|v\rangle\langle w|) = B(v)^\dagger B(w)$$

is positive, by a positive rank-one decomposition, and rotation equivariant. It agrees with its input on the k-particle sector and vanishes on smaller sectors.

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Let $V_Q \subset \bigwedge^2 U_Q$ be the spin-$Q-1$ summand, with normalized lowering basis v_p, $0 \le p \le 2Q-2$, starting at $v_0 = -e_0 \wedge e_1$. Set $B_p = B(v_p)$ and $d_a = 2Q-1$. Then

$$H_Q = \sum_p B_p^\dagger B_p$$

restricts to $H_{N,Q}$. Indeed contraction against a normalized pair in two specified tensor positions is $B(v)/\sqrt{\binom{N}{2}}$; summing over the $\binom{N}{2}$ unordered positions cancels this normalization.

Let $W_3 : V_Q \otimes U_Q \to \bigwedge^3 U_Q$ and $W_4 : V_Q \otimes V_Q \to \bigwedge^4 U_Q$ be exterior multiplication. Normal ordering gives

$$H_Q^2 = H_Q + \mathcal{L}_3(W_3(W_3^\dagger W_3 - I)W_3^\dagger) + \mathcal{L}_4(W_4 W_4^\dagger). \tag{1}$$

There are only two-, three-, and four-body terms. The zero-contraction term is $\sum_{p,q} B(v_p \wedge v_q)^\dagger B(v_p \wedge v_q)$. On two particles H_Q is the pair projection, so its square gives the two-body coefficient H_Q. On three particles $H_Q = W_3 W_3^\dagger$; subtracting that two-body term gives the three-body term displayed. Successive restrictions determine all normal-ordered coefficients, proving (1) on $\mathcal{F}_Q$.

For the falling factorial $(s)_p = s(s-1)\cdots(s-p+1)$, lowering and normalizing by $\sqrt{(2Q-2)_p p!}$ yields, for $x < y$,

$$v_p(x,y) = (x-y)\sqrt{\frac{(Q)_x (Q)_y p!}{Q(2Q-2)_p x! y!}}\mathbf{1}_{x+y=p+1}. \tag{2}$$

The norm follows also by extracting the coefficient of t^{p+1} in $\sum_{x,y}(x-y)^2\binom{Q}{x}\binom{Q}{y}t^{x+y} = 2Qt(1+t)^{2Q-2}$ and dividing by two. For fixed labels its limit is

$$v_p^*(x,y) = (x-y)\sqrt{\frac{p!}{2^p x! y!}}\mathbf{1}_{x+y=p+1}. \tag{3}$$

Here and below a superscript $*$ on a coefficient or vector denotes its limit; operator adjoints are denoted by $\dagger$.

For $Q \ge 2$, $V_Q \otimes U_Q$ is the multiplicity-free sum with spin $3Q/2 - 1 - z$, $0 \le z \le Q$. Denote the projections by $R_{3,z}$ and their dimensions by $d_{3,z} = 3Q - 1 - 2z$. The highest vector $\xi_z = \sum_{p=0}^z s_p v_p \otimes e_{z-p}$ satisfies

$$\frac{s_{p+1}}{s_p} = -\sqrt{\frac{(z-p)(Q-z+p+1)}{(p+1)(2Q-2-p)}}. \tag{4}$$

There is one such vector at each deficit; the dimensions sum to $(Q+1)(2Q-1)$, exhausting the tensor product. The exact Gram identity is

$$W_3^\dagger W_3 - I = \sum_{z=0}^{Q} q_z(Q) R_{3,z}, \qquad q_z(Q) = (-1)^z \frac{(Q)_z}{(2Q-2)_z}\left(3z - 1 - \frac{z(z+1)}{Q}\right). \tag{5}$$

For completeness, embed the tensor product in $U_Q^{\otimes 3}$, antisymmetric in its first two positions. If J is this isometry and A_3 is full antisymmetrization, then W_3 corresponds to $\sqrt{3}A_3 J$ and $W_3^\dagger W_3 = I - 2J^\dagger P_{23} J$. The two compressed swaps agree, and the compressed three-cycles have their negatives, which proves this identity directly from the six terms of A_3. Write b_z for the scalar of the compressed swap. Testing against $v_0 \otimes e_z$ gives, for $z \ge 1$,

$$2s_0 b_z = s_{z-1}\sqrt{\frac{z(Q)_z}{Q(2Q-2)_{z-1}}} - s_z(z-1)\sqrt{\frac{(Q)_z}{(2Q-2)_z}}.$$

Iteration of (4) gives

$$\frac{s_z}{s_0}=(-1)^z\sqrt{\frac{(Q)_z}{(2Q-2)_z}},\qquad \frac{s_{z-1}}{s_0}=(-1)^{z-1}\sqrt{\frac{z(Q-1)_{z-1}}{(2Q-2)_{z-1}}}.$$

Substitution yields $q_z=-2b_z$ in (5); directly $b_0=1/2$. In particular $q_z=-1$ for $z=0,1,3$ when present, so $W_3R_{3,z}=0$ there. Also

$$q_z(Q)\longrightarrow q_z^{\infty}=(3z-1)(-1/2)^z. \tag{6}$$

We use the following fixed-deficit coupling coefficients. If $j_2/(j_1+j_2)\to u^2$ and $v=\sqrt{1-u^2}$, the normalized coupled vector of deficit z and total lowering deficit T has limiting coefficients $s^{(u)}_{z,T,p}$ defined by

$$\frac{(uX-vY)^z(vX+uY)^{T-z}}{\sqrt{z!(T-z)!}}=\sum_p s^{(u)}_{z,T,p}\frac{X^pY^{T-p}}{\sqrt{p!(T-p)!}}. \tag{7}$$

The adjacent ratio in the highest-weight raising equation is

$$-\sqrt{\frac{(z-p)(2j_2-z+p+1)}{(p+1)(2j_1-p)}},$$

giving the first factor after a compatible choice of phase. Normalized lowering has denominator $\sqrt{(T-z+1)(2(j_1+j_2-z)-(T-z))}$ and tends to multiplication by $(vX+uY)/\sqrt{T-z+1}$. Induction gives (7); its orthogonal change of variables preserves the normalized-monomial inner product.

3. The exact finite certificate

Take $(p_{\max},j_{\max},i_{\max},t_{\max})=(17,18,18,17)$ and

$$S_t=\{(p,j):0\le p\le 17,\ 0\le j\le 18,\ 0\le p+j-t\le 18\}.$$

Thus $T_3=35$ and $T_4=53$. The file `code/certificate-improved.json` supplies all 64 integer rows. For a row ρ with shift t and integers ℓ,a_{pj}, use $P=10^{23}$ and set

$$\lambda=\frac{\ell}{P},\quad \alpha_{pj}=\frac{a_{pj}}{P}\sqrt{\frac{2^{p-t}t!(p+j-t)!}{p!j!}},\quad F_\rho=\lambda B_t+\sum_{(p,j)\in S_t}\alpha_{pj}c^\dagger_{p+j-t}c_jB_p. \tag{8}$$

All radicals are positive; unlisted entries are zero. Starting with Gao's published rows of denominator 10^{13}, multiply ℓ by $10^{10}-1$ and every a_{pj} by 10^{10}. This leaves α unchanged and multiplies every λ by $\kappa=9999999999/10000000000$. The four-body allowances are unchanged:

D	13	14	15	17	20	37	39
ε_D	$\frac{20271}{4\cdot10^{11}}$	$\frac{39507}{2\cdot10^{12}}$	$\frac{866443}{10^{13}}$	$\frac{9649}{125\cdot10^{9}}$	$\frac{72481}{5\cdot10^{12}}$	$\frac{129377}{10^{13}}$	$\frac{56781}{10^{13}}$

and are zero at all other D.

Put $\eta=\sum_\rho\lambda_\rho^2$ and abbreviate $s=s^{(1/\sqrt{3})}$ from (7). Define

$$e_z=\sum_\rho\sum_{T=\max(z,t_\rho)}^{35}\left[2\lambda_\rho s_{z,T,t_\rho}\sum_{p+j=T}\alpha^\rho_{pj}s_{z,T,p}+\left(\sum_{p+j=T}\alpha^\rho_{pj}s_{z,T,p}\right)^2\right]. \tag{9}$$

For $1\le D\le 53$, let $\mathcal{R}_D$ be the odd integers in $[1,D]$, and $\chi_{rs}=\mathbf{1}_{r=s=1}$. Two successive couplings give

$$S_r(D,T;p,j,k)=\sqrt{2}\,s^{(1/\sqrt{2})}_{r,j+k,j}s^{(1/\sqrt{2})}_{D-r,T-r,p}, \tag{10}$$

zero for inadmissible indices, and antisymmetric in j,k. Write $i=p+j-t$, $k=q+l-t$, $T=p+j+k$, and define

$$(E_D)_{rs}=-\sum_{\rho}\sum_{\substack{(p,j),(q,l)\in S_{t_\rho}\\ T\geq D}}\alpha^{\rho}_{pj}\alpha^{\rho}_{ql}S_r(D,T;p,j,k)S_s(D,T;q,l,i), \tag{11}$$

$$w^*_{Dr}=\sum_{\substack{p+j+k=D\\ j<k}}S_r(D,D;p,j,k)v^*_p\wedge e_j\wedge e_k,\qquad G_D=(W^*_D)^{\dagger}W^*_D. \tag{12}$$

Here W^*_D has columns w^*_{Dr}; the star is a limit label. The finite certificate asserts precisely

$$e_z<\tfrac{3}{2}q_z^{\infty}\quad(z=2,4,5,\dots,35),\qquad G_D(2\chi-E_D+\varepsilon_D I)G_D\succeq 0\quad(1\leq D\leq 53). \tag{13}$$

Appendix A gives the rational formulas that connect these inequalities to the supplied integer checker. Every one of the 33 scalar comparisons and 53 matrix checks passes. These finite checks are only one part of the proof: the next sections supply the transfer and tail estimates on the full Fock space.

4. Rotation-averaged comparison

With Haar probability measure dg, set $K_Q=d_a\int\sum_\rho F_\rho(g)^{\dagger}F_\rho(g)\,dg\succeq 0$. For one row in (8), normal ordering gives the two-body term $\lambda^2B_t^{\dagger}B_t$, the three-body terms

$$\sum_{p,j}\lambda\alpha_{pj}[(c_iB_t)^{\dagger}c_jB_p+(c_jB_p)^{\dagger}c_iB_t]+\sum_{p,j,q,l}\alpha_{pj}\alpha_{ql}\delta_{ik}(c_jB_p)^{\dagger}c_lB_q,$$

and the four-body term

$$-\sum_{p,j,q,l}\alpha_{pj}\alpha_{ql}B(v_p\wedge e_j\wedge e_k)^{\dagger}B(v_q\wedge e_l\wedge e_i).$$

These follow by replacing $c_ic_k^{\dagger}$ by $\delta_{ik}-c_k^{\dagger}c_i$. Consequently $K_Q=A_2+A_3+A_4$ with $A_2=\eta H_Q$. Schur averaging the three-body coefficient gives

$$A_3=\mathcal{L}_3\left(W_3\sum_{z\leq 35}\frac{d_a}{d_{3,z}}e_z^QR_{3,z}W_3^{\dagger}\right),\qquad e_z^Q\to e_z.$$

The convergence follows from (7). Since $d_{3,z}/d_a\to 3/2$, the strict inequalities in (13) imply, for all sufficiently large Q,

$$A_3\preceq\mathcal{L}_3\left(W_3\sum_{z\leq 35}q_z(Q)R_{3,z}W_3^{\dagger}\right). \tag{14}$$

The omitted $z=0,1,3$ vanish exactly after W_3.

For four bodies use $\widehat{W}:V_Q\otimes\bigwedge^2U_Q\to\bigwedge^4U_Q$. The second factor has pair spins $Q-r$, r odd. For fixed deficit D and large Q, the copies of spin $2Q-1-D$ are indexed by $r\in\mathcal{R}_D$ and have dimension $d_D=4Q-1-2D$. Choose normalized coupled descendants

$$|D,T;r\rangle_Q=\sum_{\substack{p+j+k=T\\ j<k}}S_r^Q(D,T;p,j,k)v_p\otimes(e_j\wedge e_k),$$

with coefficientwise limit (10), and put $w^Q_{Dr}=\widehat{W}|D,D;r\rangle_Q$. Schur averaging, including the matrices between equivalent copies, gives

$$A_4=\mathcal{L}_4\left(\widehat{W}\bigoplus_{D=1}^{53}\frac{d_a}{d_D}(I_{d_D}\otimes E_D^Q)\widehat{W}^{\dagger}\right),\qquad E_D^Q\to E_D. \tag{15}$$

The target $\mathcal{L}_4(W_4W_4^\dagger)$ has copy matrix χ: its domain is exactly the $r=1$ copy. In the units of (15) this is $(d_D/d_a)\chi$, tending to 2χ. At even D that copy wedges to zero by symmetry. Other target spin blocks are positive and can be discarded. No linear independence of the wedged columns is needed.

5. Relative transfer in the presence of Gram kernels

For any finite W and Hermitian C, with $G = W^\dagger W$,

$$GCG \succeq 0 \quad \Longleftrightarrow \quad WCW^\dagger \succeq 0. \tag{16}$$

Both test the quadratic form of C on $\operatorname{ran} G = \operatorname{ran} W^\dagger$. Thus (13) proves positivity of the limiting four-body form and of its lift, without asserting positivity of the uncompressed coefficient matrix.

All highest-vector operators needed here use only modes $0,\dots,54$. First work on $\mathcal{F}(\mathbb{C}^{55})$. For $Q \geq 54$, on an occupation set A define

$$\Delta_Q|A\rangle = \prod_{x\in A} d_Q(x)|A\rangle, \qquad d_Q(x) = \sqrt{(Q)_x/Q^x}, \qquad r_p(Q) = \sqrt{(2Q)^p/(2Q-2)_p} \geq 1.$$

Then $0 < \Delta_Q \leq I$ and $\Delta_Q^{\pm 1} \to I$ in norm on this fixed finite space. Tracking the removed occupation labels in (2)–(3) gives the exact identities

$$B_p = r_p(Q)\Delta_Q^{-1} B(v_p^*)\Delta_Q, \tag{17}$$

$$B(v_p \wedge e_j \wedge e_k) = \frac{r_p(Q)}{d_Q(j)d_Q(k)} \Delta_Q^{-1} c_k c_j B(v_p^*)\Delta_Q.$$

The inverse diagonal acts on the output sector; repeated labels make both sides zero. With $h_* = \sum_{p=0}^{53} B(v_p^*)^\dagger B(v_p^*)$ and $h_Q = \sum_{p=0}^{53} B_p^\dagger B_p$, these identities imply

$$\Delta_Q h_* \Delta_Q \preceq h_Q, \tag{18}$$

because $r_p \geq 1$ and $\Delta_Q^{-1} \geq I$.

Fix D, and index $b = (p,j,k)$ with $p+j+k = D$, $j<k$. Let $A_b = c_k c_j B(v_p^*)$. Since at fixed p there are $\lfloor (D-p+1)/2 \rfloor$ choices,

$$\sum_b A_b^\dagger A_b \preceq c_D h_*, \qquad c_D = \lceil D/2 \rceil.$$

Put $C_D^Q = (d_D/d_a)\chi - E_D^Q + \varepsilon_D I$. By (17),

$$B(w_{Dr}^Q) = \Delta_Q^{-1} \sum_b \beta_{rb}^Q A_b \Delta_Q, \qquad \beta_{rb}^Q = S_r^Q(D,D;p,j,k)\frac{r_p(Q)}{d_Q(j)d_Q(k)} \longrightarrow S_r(D,D;p,j,k).$$

The Hermitian matrices $M_D^Q = (\beta_D^Q)^\dagger C_D^Q \beta_D^Q$ therefore converge to $M_D^* = (\beta_D^*)^\dagger(2\chi - E_D + \varepsilon_D I)\beta_D^*$. By (16) and positivity of the lift, $\sum_{b,c}(M_D^*)_{bc}A_b^\dagger A_c \succeq 0$.

Here is the explicit relative error used in the accepted proof. For every occupation set, the product inequality $\prod_i(1-a_i) \geq 1 - \sum_i a_i$ for $0 \leq a_i \leq 1$ gives

$$\prod_{x\in A} d_Q(x)^2 \geq 1 - S/Q, \qquad S = \sum_{x=0}^{54}\sum_{k=0}^{x-1} k = \binom{55}{3} = 26235.$$

For $Q > S$, therefore, $\|\Delta_Q^{-2}\| \leq Q/(Q-S)$ and $\|\Delta_Q^{-2} - I\| \leq S/(Q-S)$. Write $\delta_D(Q) = \|M_D^Q - M_D^*\| \to 0$ and $\mu_D = \|M_D^*\|$. Direct multiplication gives

$$\sum_{r,s}(C_D^Q)_{rs} B(w_{Dr}^Q)^\dagger B(w_{Ds}^Q) = \Delta_Q \sum_{b,c}(M_D^Q)_{bc} A_b^\dagger \Delta_Q^{-2} A_c \Delta_Q.$$

Subtracting the positive limiting compressed form leaves a block operator with norm at most $(Q\delta_D(Q)+S\mu_D)/(Q-S)$. Applying this bound to $(A_b\Delta_Q\psi)_b$ and using (18) proves

$$\sum_{r,s}(C_D^Q)_{rs}B(w_{Dr}^Q)^\dagger B(w_{Ds}^Q)\succeq -c_D\frac{Q\delta_D(Q)+S\mu_D}{Q-S}h_Q. \tag{19}$$

This argument uses compressed positivity only, and makes no assumption about stability of Gram kernels.

Order the 55 local modes before the spectator modes. Under the Fock factorization, every local annihilator is its local operator tensored with the identity; parity factors are needed only for later-mode annihilators, which do not occur. Thus (19) tensors with the identity and holds uniformly in total particle number. Conjugation by rotations and integration preserve order. Schur's lemma gives $d_a\int h_Q(g)\,dg = 54H_Q$. Consequently a total error is

$$\epsilon_Q^{\mathrm{loc}} = 54\sum_{D=1}^{53} c_D\frac{Q\delta_D(Q)+S\mu_D}{Q-S}\longrightarrow 0. \tag{20}$$

All dimensions and cutoffs in this estimate are fixed independently of particle number. The condition $Q>S$ is a domain for this estimate, not a claimed final gap threshold.

6. Allowance and tail estimates

Put $X_D = d_a\int\sum_r B(w_{Dr}^Q(g))^\dagger B(w_{Dr}^Q(g))\,dg\succeq 0$. Averaging (19), summing, and using (15) yields

$$\mathcal{L}_4(W_4W_4^\dagger)-A_4+\sum_D\varepsilon_D X_D\succeq -\epsilon_Q^{\mathrm{loc}}H_Q. \tag{21}$$

For $1\le T\le 53$ and $Q>T+1$, the vectors $v_p\otimes(e_j\wedge e_k)$ with $p+j+k=T$, $j<k$ are an orthonormal weight basis of size $n_T=\lfloor (T+1)^2/4\rfloor$. The normalized coupled descendants $|D,T;r\rangle_Q$, $D\le T$, form another such basis; $\sum_{D\le T}\lceil D/2\rceil = n_T$. Averaging the rank-one projectors of either basis and lifting after $\widehat{W}$ gives

$$\sum_{D\le T}X_D = d_a\sum_{\substack{p+j+k=T\\ j<k}}\int B(v_p(g)\wedge e_j(g)\wedge e_k(g))^\dagger B(v_p(g)\wedge e_j(g)\wedge e_k(g))\,dg\preceq n_T H_Q. \tag{22}$$

The inequality uses $\|c_kc_j\|\le 1$ and $d_a\int B_p(g)^\dagger B_p(g)\,dg = H_Q$. Let $\bar\varepsilon_D=\max_{D'\ge D}\varepsilon_{D'}$, $\bar\varepsilon_{54}=0$, and $\omega_T=\bar\varepsilon_T-\bar\varepsilon_{T+1}\ge 0$. Then (22) implies

$$\sum_D\varepsilon_D X_D\preceq\sum_T\omega_T n_T H_Q=\pi_S H_Q,\quad \pi_S=\sum_{D=1}^{53}\lceil D/2\rceil\bar\varepsilon_D=\frac{3358343}{312500000000}. \tag{23}$$

For the three-body tail, let $x=a\otimes b$ be a product of unit vectors in $V_Q\otimes U_Q$, with $\theta=\langle x,R_{3,z}x\rangle>0$. Multiplicity-free Schur averaging gives $\int|x(g)\rangle\langle x(g)|\,dg\succeq(\theta/d_{3,z})R_{3,z}$. Lift after W_3 and use $\|B(b)\|\le 1$, then average on V_Q, to obtain

$$\mathcal{L}_3(W_3R_{3,z}W_3^\dagger)\preceq\frac{d_{3,z}}{d_a\theta}H_Q. \tag{24}$$

For $x=v_p\otimes e_{z-p}$, its overlap is $s_p(Q)^2$ from (4), so take $c_z(Q)=1/\max_p s_p(Q)^2\le z+1$. For fixed z, the limit of these probabilities is $\binom{z}{p}(1/3)^p(2/3)^{z-p}$ by (7). Comparing adjacent masses gives a mode $p_z=\lfloor(z+1)/3\rfloor$, hence

$$c_z(Q)\longrightarrow c_z=\frac{3^z}{\binom{z}{p_z}2^{z-p_z}}. \tag{25}$$

The maximizing index is chosen at finite Q before taking the limit.

Use $c_z(Q)$ for odd $37 \le z \le 301$ and $z+1$ for larger odd z. The bracket in (5) is at least $2z-2>0$ for $2 \le z \le Q$, so only odd blocks are negative. For $Q \ge 4$, the successive factors in $(Q)_z/(2Q-2)_z$ decrease and are at most $2/3$. Also $d_{3,z}/d_a \le 2$ and $|q_z(Q)| \le 3z(2/3)^z$. The relative tail summands from (24) are dominated by $6z(z+1)(2/3)^z$, a summable sequence. Define summands with $z>Q$ to be zero; no out-of-range factorial is evaluated. Dominated convergence therefore gives, uniformly in particle number,

$$\mathcal{L}_3\left(W_3 \sum_{z=36}^{Q} q_z(Q) R_{3,z} W_3^\dagger\right) \succeq -\delta_Q H_Q, \qquad \delta_Q \longrightarrow h+R, \tag{26}$$

where both constants are rational:

$$h = \sum_{z=37,39,\dots,301} \frac{3(3z-1)3^z}{2^{2z+1-p_z}\binom{z}{p_z}}, \qquad R = \frac{69313}{2^{300}}. \tag{27}$$

Indeed $R = (3/2)\sum_{z\ge 303,\ z\ \mathrm{odd}} (3z-1)(z+1)2^{-z}$. Set $z = 303+2m$; the polynomial becomes $276032 + 3640m + 12m^2$. Using $\sum 4^{-m} = 4/3$, $\sum m4^{-m} = 4/9$, and $\sum m^2 4^{-m} = 20/27$ gives exactly the displayed R.

7. Exact margin and the physical gap

The old pair cost is $\eta_{\mathrm{old}} = 390486114442087984799240 01/(5\cdot 10^{25})$; the modified rows have $\eta = \kappa^2 \eta_{\mathrm{old}}$. Define

$$\Gamma = 1 - \kappa^2 \frac{390486114442087984799240 01}{5\cdot 10^{25}} - \frac{3358343}{312500000000} - h - \frac{69313}{2^{300}}. \tag{28}$$

Exact integer cross multiplication, reproduced by the supplied script, gives

$$\Gamma - \gamma > \frac{2959}{10^{16}} > 0, \qquad \gamma - 0.2190037845063196 = \frac{33070359201}{2500000000000000} > 0. \tag{29}$$

No rounded decimal enters these comparisons.

Combining (1), (14), (21)–(23), and (26), with $K_Q = A_2 + A_3 + A_4 \succeq 0$, yields on the entire $\mathcal{F}_Q$

$$H_Q^2 \succeq K_Q + (1-\eta-\delta_Q-\pi_S-\epsilon_Q^{\mathrm{loc}})H_Q \succeq (\Gamma - e_Q)H_Q, \quad e_Q = \epsilon_Q^{\mathrm{loc}} + |\delta_Q - h - R| \to 0. \tag{30}$$

The strict finite low-block comparisons, the fixed local relative error, and the dominated scalar tail all have thresholds independent of particle number. Choose an integer $Q_0 > 26235$ beyond the low-block thresholds and such that $e_Q < \Gamma - \gamma$ for every integer $Q \ge Q_0$. This exists by convergence. The spectral theorem then gives $H_{N,Q} \succeq \gamma(I - P_{\ker H_{N,Q}})$ simultaneously in every particle sector for $Q \ge Q_0$.

It remains to identify the kernel at $Q = 3(N-1)$. Realize U_Q as degree-Q homogeneous spinor polynomials and write $b_{ij} = u_i v_j - u_j v_i$. The pair spin-$Q-r$ summand has highest polynomial $b_{ij}^r u_i^{Q-r} u_j^{Q-r}$, whose descendants remain divisible by b_{ij}^r. Antisymmetric pair sectors have odd r. A vector in $\ker H_{N,Q}$ is killed by every positive pair projector, so every b_{ij}^3 divides its polynomial. Each bracket is primitive and linear in u_i over the fraction field of the other variables, hence irreducible by Gauss's lemma. Distinct unordered pairs give nonassociate brackets. Unique factorization implies divisibility by $\prod_{i<j} b_{ij}^3$. Its degree already exhausts Q in each particle variable, so the quotient is constant. Conversely this product is antisymmetric and, after factoring b_{ij}^3, its maximum pair spin is $Q-3$; multiplication by the invariant bracket introduces no spin $Q-1$. Every pair projector therefore kills it. The kernel is the Laughlin line. Taking $N_0 = \max(2, 1+\lceil Q_0/3\rceil)$ proves Theorem 1.

8. Improvement over the original uniform gap

The original spherical gap is $1/25 = 0.04$ [2]. The exact certificate here yields

$$\gamma = 0.21901701265 > 0.04, \qquad \gamma/(1/25) = 5.47542531625.$$

Both statements use coefficient one per pair projector and flux $Q = 3(N-1)$ for the unperturbed spherical fermionic V_1 Hamiltonian, for all sufficiently large N. The proof retains OpenAI's Fock-space comparison and Gao's enlarged certificate, product-vector averaging and staircase allowances [2, 1]. The row adjustment is specified in Section 3.

Appendix A. Rational certificate checks

The following formulas specify the exact correspondence used by `code/check_certificate.py`; they are part of the accepted proof. Let

$$U(c,z,T,p) = [x^p](x-1)^z(cx+1)^{T-z} = \sum_a (-1)^{z-a}\binom{z}{a}\binom{T-z}{p-a}c^{p-a},$$

zero for inadmissible indices. Expansion of (7) gives

$$s_{z,T,p} = U(2,z,T,p)\sqrt{\frac{2^z\binom{T}{z}}{3^T 2^p\binom{T}{p}}}.$$

For $X_{\rho,T} = \sum_{p+j=T} a^\rho_{pj} U(2,z,T,p)$, equation (9) becomes the rational formula

$$P^2 e_z = \sum_{\rho,T} \frac{2^z\binom{T}{z}}{3^T 2^{t_\rho}\binom{T}{t_\rho}} X_{\rho,T}\bigl(X_{\rho,T} + 2\ell_\rho U(2,z,T,t_\rho)\bigr). \tag{31}$$

For four bodies put $u_r = 1/(r!(D-r)!)$ and $V_r(D,T;p,j,k) = U(1,r,j+k,j)U(1,D-r,T-r,p)$. Write $(E_D)_{rs} = \sqrt{u_r u_s}Y_{rs}$ and $(G_D)_{rs} = \sqrt{u_r u_s}Z_{rs}$. Then

$$S_r = V_r\sqrt{2^{1-(j+k)-T+r} j!k!p!u_r/(T-D)!},$$

and all radicals cancel in

$$P^2 Y_{rs} = -\sum_{\rho,p,j,q,l} a^\rho_{pj}a^\rho_{ql}V_r(D,T;p,j,k)V_s(D,T;q,l,i)2^{1-2T+p+q-t+(r+s)/2}\frac{i!k!t!}{(T-D)!}, \tag{32}$$

with the restrictions of (11). Ordered entry pairs are intentional. For an increasing quadruple A of total sum $D+1$, put $f(A) = \prod_{x\in A} x!$ and

$$L_r(A) = 2^{(1-2D+r)/2} \sum_{\substack{x<y\in A\\ j<k\in A\setminus\{x,y\}}} \sigma(x,y,j,k)(x-y)p!j!k!V_r(D,D;p,j,k), \qquad p = x+y-1.$$

Here σ is the sign for reordering (x,y,j,k) increasingly. The coefficient of w^*_{Dr} on A equals

$$\sqrt{u_r/f(A)}L_r(A), \qquad Z_{rs} = \sum_A L_r(A)L_s(A)/f(A).$$

Since r,s are odd, the powers of two in these rational formulas have integer exponents. With $D_u = \mathrm{diag}(u_r)$, the second matrix in (13) is congruent by a positive diagonal to

$$M_D = Z_D B_D Z_D, \qquad B_D = \mathrm{diag}\bigl((2\mathbf{1}_{r=1} + \varepsilon_D)u_r\bigr) - D_u Y_D D_u.$$

The checker factors the sums in (32) by i,k, with $T = t+i+k$. It clears denominators by $P^2 2^{2T_4+t_{\max}+2}(T_4-D)!$ and uses the integer multinomial $(D+1)!/f(A)$ for Z_D. These are positive scalings; dividing entries by their positive greatest common divisor also preserves signs.

Positivity is tested by exact symmetric Schur elimination. A negative diagonal fails. If all diagonals vanish, positivity requires all entries to vanish, by testing $e_i + te_j$. Otherwise a positive pivot p splits off by congruence; replace the remaining block by $pA_{\mathrm{rest}} - A_{\mathrm{column}}A_{\mathrm{row}}$ and repeat. This is a positive multiple of the Schur complement and handles exact kernels. All blocks pass, with ranks for $D = 1, \ldots, 53$:

$$0, 0, 0, 0, 1, 0, 1, 1, 2, 1, 3, 2, 4, 3, 5, 4, 6, 5, 7, 6, 8, 7, 9, 8, 10, 9, 11,$$
$$10, 12, 11, 13, 12, 14, 13, 15, 14, 16, 14, 17, 13, 13, 11, 11, 9, 9, 7, 7, 5, 5, 3, 3, 1, 1.$$

The separately supplied Gao verifier tests the characteristic coefficients: if every coefficient of $\det(xI + M_D)$ is nonnegative, it is positive for $x > 0$, so a symmetric M_D has no negative eigenvalue. Both tests use exact arithmetic. Reproduction commands, certificate hashes, citation provenance, and compact validation results are supplied with the packet; raw computational logs are excluded.

References

1. Qiang Gao, *Improving the lower bound on the Laughlin gap: An SDP search*, arXiv:2610.12700v1 [cond-mat.str-el], 2026, October 8, 2026.
2. OpenAI, *A Fock-space inequality and the Laughlin spectral gap*, OpenAI Math Release preprint, 2026, Dated September 24, 2026; source at commit `1581d4c`.
3. ———, *Uniform Stability of the Spherical Laughlin Gap*, OpenAI Math Release preprint, 2026, Dated October 5, 2026; source at commit `1581d4c`.

Mathematical supplement

The proof supplements, exact certificate data and arithmetic programs referenced by filename are included in the attached mathematical supplement (ZIP). Open the attachment using a PDF reader with file-attachment support.